

A Machine Learning Based Approach for Wine Quality Prediction

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Abstract- Wine is a popular drink across the globe and the gender. Older the Wine, better is the taste but, expensive. The Wine quality is measured based on the important parameters, such as free Sulphur dioxide, Volatile acidity, Citric Acid and Residual sugar. The traditional way of Wine quality assessment was time consuming. This paper gives an automatic prediction of Wine quality, as good or bad, using machine learning approaches which are Neural Networks, Logistic Regression and Support Vector Machine are implemented on standard datasets of Portuguese "Vinho Verde" Wine. The results are compared with standard values. The support vector Machine has achieved superior than other techniques with error of 0.003. The quality rate for SVM is 7.99. The work is useful in Wine industry for quality testing and assurance for customers.

Keywords: Machine Learning, Prediction, Support Vector Machine, Logistic Regression.

I. INTRODUCTION

Wine is one of the popular drinks in the world. Wine quality depends on the features used in manufacturing of a Wine. Its quality is reliant on stages, commonly older the Wine, better the taste. Quality of the Wine is vital for customers, and also for manufacturers in the current modest marketplace to increase the production. All Wines can be categorised into five primitive types as Sparkling Wine, Red wine, Rose wine, White Wine and Dessert Wine. Black Grapes are used in the preparation of Red Wine. The colour of red Wine is usually light to dark. The Green and Black Grapes are used in production of white Wine. The Red and white Wines appear as shown in Figure 1. If the Wine quality is not good, then the numerous techniques have to be applied from the start to improve its quality. The persons have their individual views about the Wine quality, mainly based on the taste. The consideration of person's taste in the classification of wine quality is quite interesting. The improvement in various technologies, the producers in production can trust on numerous strategies for checking the quality of Wine. Then the producers can get better knowledge of Wine quality [09].

Nowadays, all manufacturers employing different practises to take full advantage of production and constructing the entire practice well-organized. These procedures are flattering luxurious with period, and their difficulties are also growing. The 11th major Wine manufacturer is Portugal in the world. Documentation and assessment of Wine are vital rudiments in Portugal's Wine manufacturing. In earlier times, when there was a deficiency of technology and machinery, the analysis and quality proclamation of Wines cannot be attained.

A. Red Wine and White Wine

The dark-coloured grape varieties are used to prepare the Red Wine. The colour of the Wine could be deep violet, brown for older red Wines. The juice from maximum purple grapes is greenish-white, the red colour approaching from anthocyanin substances contained in the peel of the grape [5]. Most of the red Wine manufacture procedure includes mining of colour and essence constituents from the grape skin. Without the skin contact of grapes, wine can be fermented. The colour of white Wine could yellow-gold, straw-yellow, yellow-green. The alcoholic fermentation of non-coloured grapes can be used in preparation of white Wine. White wine can sustain till 4,000 years with its better taste and colour. [6].



Fig. 1. Sample of Red wine and White wine

The different Manufactures of Wine in terms of production are given in Table 1. The major comprehensive wine marketplaces are the USA, China, and France. Due to their higher populations, while the major traders of wine are United Kingdom, Germany and the USA, where wine making is lesser than intake.

TABLE I. LEADING WINE MANUFACTURERS WORLDWIDE

Sl.N O	Country	Manufactures	Production Litres/ year
1	India	Sula Vineyards ,Chateau Indage Limited, Grover Vineyards Limited	1,76,00,000
2	Italy	Bosco Viticultor, Casa Girelli. Cantina Cortecchia,	4,250,000
3	France	Domaine Virgile Joly, Domaine Cazes, Domaine de	3,641,900

4	Spain	Bodegas Torres, Marqués de Riscal, Dinastia Vivanco, Bodegas Protos,	3,248,000
5	United States	Constellation Brand, E&J Gallo, The Wine Group	2,333,900
6	Australia	Barossa Valley Estate, Brokenwood Wines, Cape Mentelle	1,369,000
7	Argentina	Achaval-Ferrer, Bodega Catena Zapata, Bodega Chacra, Bodega	1,182,100
8	China	Bate Lu Wine, Changyu Pioneer Wine, Chateau Aroma, Conghua	1,163,600
9	South Africa	Mountain Ridge, Stellenberg Wines, Cape Wine Company, Sarah's Creek	1,080,100
10	Chile	Merlot, Carménère, Zinfandel, Petite Sirah, Sangiovese Cabernet franc.	949,200
11	Germany	Riesling, Spätburgunder (Pinot Noir)	746,200

Wine Companies in India are growing over the last few years. Maharashtra, Karnataka, Andhra Pradesh, Himachal Pradesh are some of the major states of India contributing more in the growth of Wine manufacturing and long-winded in the production of wines.

Consumption of red Wine in control may have few health aids, containing improving heart related problems, gut, and mind health.[11] Since it comprises composites with antioxidant, anti-inflammatory and lipid-improving properties. Wine can be used in preparation of chocolate, Cosmetics like perfume, cleansing products etc..[12]. The parameters which play major role in deciding the quality of wine are Stable acidity, Total Sulphur Dioxide, Volatile acidity, Residue Sugar, Chlorides, Citrus Acid Free Sulphur dioxide, Solidity, PH, Sulphates, Alcohol etc. [13].

The analysis of Wine quality dependant on chemical or other characteristics has been very challenging in the previous years due to less technical tools which consume a lot of time and money venture. Currently it can be simply categorized with the assistance of machine learning methods. Throughout this manufacturing method, one can adjust the factors that directly regulate the quality of Wine. This provides the producer an improved knowledge to change the wine quality by refinement of diverse factors which affects the Wine improvement method. Therefore, the study of the elementary factors which control the Wine quality is important.

Machine Learning could be a substitute in recognizing the maximum key factors that govern the quality of the wine. The Wine data samples are existing in numerous repositories like UCI Machine Learning Repository, Portuguese Wine Reviews Dataset from "BlogOsVinhos", Global Wine points, Texas Wineries and Kaggle etc.. The more enhancements of ML approaches and their attainment in the last few years, there have been numerous exertions while defining Wine quality with the existing documents [2].

In this research work, the data samples of both white and red wine of the Portuguese "Vinho Verde" are considered for

prediction analysis. We implemented three ML algorithms such as Support Vector Machine, Logistic Regression and Neural Networks. The feature reduction is performed by employing Neural Networks Regression analysis and Principal Component Analysis method. The Performances' of classification algorithms are compared and depicted.

The next part of the paper is structured into four sections. In section 2, we discuss the existing works on prediction of Wine and its quality, the proposed methodology is described in section 3 and in section 4 we presented Results and Discussions.

II. LITERATURE SURVEY

(Piyush Bhardwa et al. 2022) developed a machine Learning Model using the artificial data collection and existing experimental information collected from various and different areas across New Zealand. The dataset consisted of 18 Pinot Noir wine data samples with dissimilar 54 features taken in to consideration for study. Seven machine learning algorithms such as RF, SVM, DT, KNN, Ada Boost, XGBoost, GNB implemented on data samples. The highest performance is obtained from Random Forest Algorithm [1].

(K. R. Dahal, J. N. Dahal, H. Banjade, S. Gaire 2021) compared performance of some ML Techniques such as Gradient Boosting Regression (GBR), Ridge Regression (RR), multi-layer Artificial Neural Network (ANN) and Support Vector Machine (SVM) for quality prediction of wine. Many features that control the wine quality are studied. The carried out analysis demonstrate that GBR beats other Methods' performance correspondingly [2].

(Shaw, B., Suman, A.K., Chakraborty, B. 2020) developed model for prediction of wine quality using few approaches of Machine Learning. The approaches such as Multi-Layer Perceptron, Random Forest and Support Vector Machine were used for study. The highest accuracy is achieved with the RF technique is about 81.1% [3].

(Mahima, Ujjawal Gupta, Yatindra Patida 2019) designed the user friendly model forecasts the wine quality by choosing the significant factors of wine which contribute an important part in defining the wines quality. The dataset from UCI Machine Learning Repository is considered for analysis. The KNN and Random Forest methods are employed for prediction of wine quality. The output is determined on the basis of good, bad and average scale [4].

(Satyabrata Aich, Mangal Sain and Jin-Han Yoon 2019) developed model using Recursive Feature Elimination (RFE) technique for feature selection, as well as nonlinear and probabilistic classification. The comparison of different classifiers was carried out with Original Feature Sets (OFS) as well as Reduced Feature Sets (RFS). The accuracy obtained is ranging from 97.61% to 99.69% among the dissimilar feature set [5].

(Yogesh Gupta 2018) developed a model for selection of significant features for wine quality prediction. They implemented linear regression, neural network and support vector machine for predict quality in two types. The datasets of both Red Wine and White Wine are considered for study. The conclusion was made from the results that more precisely dependent variable could be predicted when only significant attributes are measured in prediction rather than measuring all attributes [6].

III. PROPOSED METHODOLOGY

It comprises of four stages as depicted in Figure 2. Stages are mentioned as a) Data collection b) Feature extraction c) Feature reduction and d) classification. Details of the stages are shown in Figure 2.

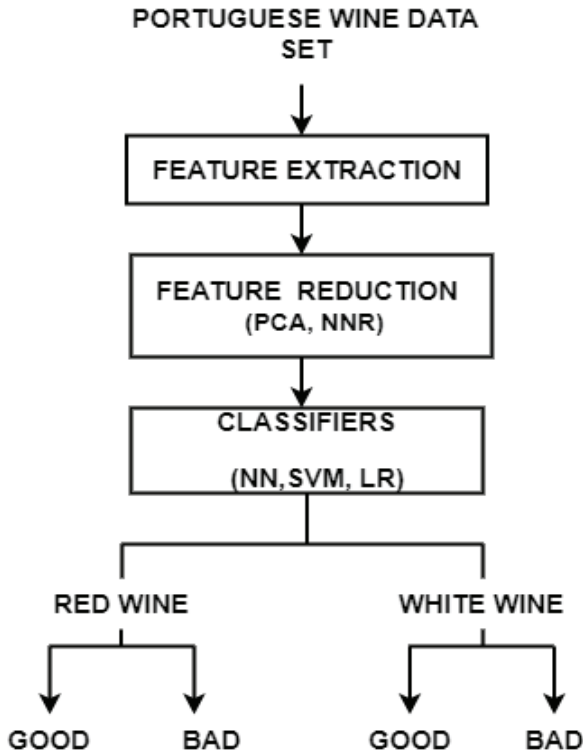


Fig. 2. Block diagram of methodology

Algorithm

Input: Data set of White and Red wines

Output: Good or Bad Quality wine.

Description: F1: Volatile acidity, F2: Citric Acid, F3: Residual sugar, F4: Free sulphur dioxide and NNRA: Neural Network Regression Analysis.

Start

Step 1: Read Red and White wine data from UCI machine learning repository.

Step 2: Obtain features manually and from feature reduction algorithms.

Step 3: Apply SVM, Logistic Regression and ANN classification algorithms.

Step 4: Compare the results against standard values.

Step 5: Report the results of best performing algorithm

Stop.

A. Data Collection

In this study, “Vinho Verde” wine from the Portuguese grape vineyard is considered. This consists of huge heap of datasets that have been extensively used by the research laboratories. Data samples are taken from UCI machine learning website. UCI Repository maintains two separate data files with respect to white and red wine. White wine

data set contains 4899 records with eleven features and Red wine contains 1600 records with same features set. Red wine is more preferred compared to white wine because of its high acceptance rate.

B. Feature Extraction

Collected wine data set has following physiochemical features: stable acidity (g[tartaric acid]/dm³), volatile acidity (g[acetic acid]/dm³), total sulfur dioxide (mg/dm³), chlorides (g[sodium chloride]/dm³), pH level, free sulfur dioxide (mg/dm³), solidity (g/cm³), residue sugar (g/dm³), citrus acid (g/dm³), sulphates (g[potassium sulphate]/dm³), and alcohol (vol%) and their respective standard values are given in Table 2.

TABLE II. MEAN AND STANDARD DEVIATION VALUES

Sl.no	Ingredients of Wine	Mean	Standard deviation
1	Stable acidity	6.954	0.943
2	Volatile acidity	0.378	0.200
3	citrus acid	0.434	0.221
4	Residue sugar	6.491	5.082
5	Chlorides	0.055	0.031
6	Free sulphur dioxide	36.30	18.00
7	Total sulphur dioxide	139.4	43.49
8	Solidity	0.996	0.003
9	PH	4.188	0.251
10	Sulphates	0.589	0.214
11	Alcohol	10.61	2.230

C. Feature Reduction

Both datasets have similar physicochemical features. Among these features only few will decide the quality of wine. To save memory and execution time, Feature reduction algorithms are employed. Neural network regression analysis algorithm and Principal component analysis (PCA) are applied for feature reduction. The list of reduced features is given in Table 3.

TABLE III. REDUCED FEATURES USING PCA

Sl.no	Ingredients of Wine
1	Volatile acidity
2	citrus acid
3	Residue sugar
4	Alcohol
5	Free SO ₂

1) Neural Network Regression Analysis (NNRA)

In the proposed work NNRA is used for feature reduction. This helps in finding best features out of eleven physiochemical features of wine those play important role in deciding quality of wine.

Standard Regression Co-efficient can be obtained using (1)

$$Z_i = \frac{X_i - \text{Mean}}{\text{Standard Deviation}} \quad (1)$$

where x_i is i th ingredient value and Mean is mean of all ingredient values of wine. Standard Deviation is calculated over all the ingredients value. The mean value is calculated using (2)

$$\text{Mean} = \frac{(X_1 + X_2 + \dots + X_n)}{n} \quad (2)$$

$X_1, \dots, X_n$ are ingredient values of wine.

These two parameters predict an output variable as a function of inputs such as to calculate mean, standard deviation and output is regression coefficient.

2) Principal Component Analysis (PCA)

In the proposed work, PCA is another method used for feature reduction, thereby helping to choose suitable features boosting for the increase of success rate of classification of data. NNRA selects only 4 features, but with PCA, 5 feature are selected i.e., alcohol is the additional feature, accounts to increase the quality of strength of taste of wine. It identifies the similarities and differences in a high dimensional data to meet standard deviation, eigen values and eigen vectors.

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{False Negative}}$$

D. Classification

Input to machine is either red wine data set or white wine dataset expected output is to obtain classified them as good quality or bad quality wine. To achieve this goal, Neural network, SVM and Logistic regression are employed. According to the survey carried out the features are selected using (3).

1) Neural Network

Neural networks can solve many interesting problems so we are using it to classify wine data. The data set must to be normalized for training of neural network. Normalization helps all data values of data set to range between 0 and 1 that can be obtained by below formula.

$$Y_n = \frac{Y - Y_{\min}}{Y_{\max} - Y_{\min}} \quad (3)$$

Y is the Original value, $Y_{\min}$ is the minimum coordinate value along Y axis obtained for each feature

$Y_{\max}$ is the maximum coordinate value along Y axis obtained for each feature

2) Support Vector Machine (SVM)

SVM is the best suited algorithm to classify the wine quality either as good or bad [5]. The quality is calculated for selected features dataset. The quality Q is defined as $Q = \{(y_i, c_i) | y_i \in P \text{ value}\}$, where Y_i is the vector along Y axis and C_i is the Decision boundary. P value is the probability value

3) Logistic Regression Algorithm

It is a supervised machine learning algorithm is applied when the dataset consists of independent variables and is used in predicting the output of categorical dependent variables. The output will be either good or bad. It is used for the classification problems and separate datasets based on continuous and discrete properties. p -value is $P(\text{Ov} \leq \text{Sf} | H_0)$, where P is probability, Sf is Standard feature values and Ov is Observed Values under the test hypothesis H_0 .

IV. RESULTS AND DISCUSSIONS

There are standard values for ingredients of wine in terms of mean and standard deviations given in Table 2. The four features that are considered are Residual sugar, volatile acidity, citrus acid and sulphur dioxide (SO_2). The neural network regression analysis is used for feature reduction. The correlation coefficient is denoted as R indicating how much quality is correlated with remaining wine quality predictors. The value of the correlation coefficient is 0.603.

TABLE IV. REGRESSION COEFFICIENTS FOR RED WINE

$R=0.6$	Standard regression coefficient	Standard error	P value
Stable acidity	0.0541	0.056	0.336
Volatile acidity	0.240	0.027	0.000
Citrus acid	-0.044	0.036	0.215
Residue sugar	0.029	0.026	0.277
Chlorides	0.111	0.024	0.000
Free sulphur dioxide	0.055	0.028	0.045
Total sulphur dioxide	0.133	0.030	0.000
Solidity	-0.042	0.050	0.400
PH	0.079	0.037	0.030
Sulphates	0.192	0.024	0.000
Alcohol	0.364	0.035	0.000

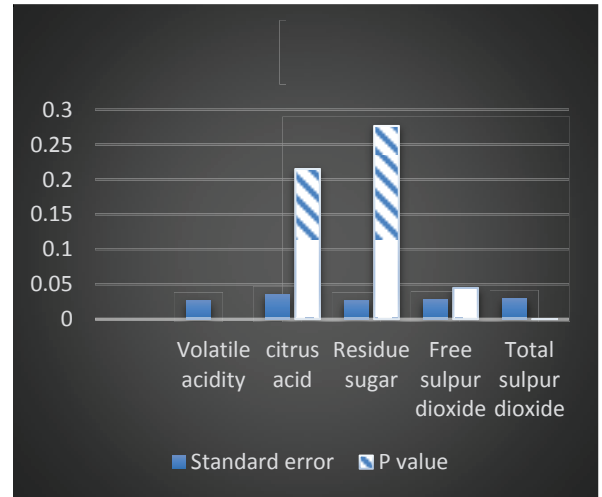


Fig. 3. Error and P values for red wine.

The mean error value and mean p value for red wine are 0.005 and 0.011 respectively are obtained from figure 3.

Table 4 depicts the overall regression summary of red wine quality and indicates the dependency variable that is quality on different predictors. The regression coefficient is

0.6 that denotes 33.71% dependency of quality on total predictors. If regression coefficient falls below 0.04 means it is completely different from zero.

The four ingredients of wine such as volatile acidity, citrus acid, residual sugar and both types of sulphur dioxide belonging to category of Free SO₂ and Total SO₂. The value of p for these contents obtained is below 0.05. The percentage obtained is 29.12% wine quality which is the dependent variable, if this value somehow falls below fifty percent, there are few ingredients in Wine that contribute to the quality of either Red Wine or White Wine prescribed in Table 3. In Table 4 and Table 5, the p value is almost less than 0.05, meaning that probability of the selected predictors is very best if compared with different predictors of Wine.

TABLE V. REGRESSION COEFFICIENT FOR WHITE WINE

R=0.5	Standard regression coefficient	Standard error	P value-
Stable acidity	0.062	0.019	0.001
Volatile acidity	0.212	0.012	0.000
citrus acid	0.003	0.013	0.817
Residue sugar	0.467	0.043	0.000000
Chlorides	-0.007	0.013	0.650
Free sulphur dioxide	0.0711	0.016	0.000
Total sulphur dioxide	-0.013	0.018	0.449
Solidity	0.507	0.064	0.000
PH	0.117	0.018	0.000
Sulphates	0.081	0.013	0.000
Alcohol	0.268	0.033	0.000

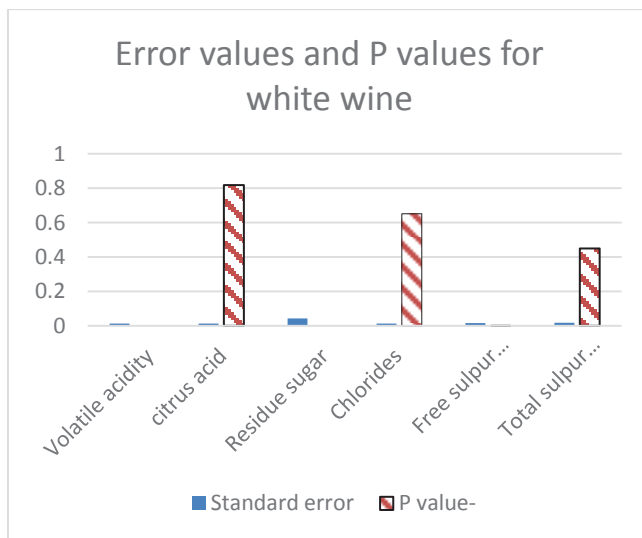


Fig. 4. Error and p values for White Wine.

The mean error value and mean P value for White Wine are 0.0033 and 0.060 respectively are obtained from figure 4. The Table 6 gives the precision and recall values obtained using neural network classifier.

TABLE VI. PCA RESULTS

Type of wine	Neural Network Classifier	
	Precision	Recall
Red Wine	64.5	64.7
White wine	64.3	64.8

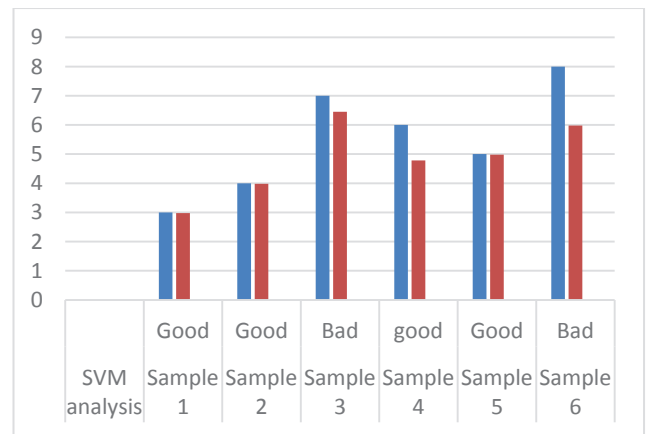


Fig. 5. Prediction using SVM analysis

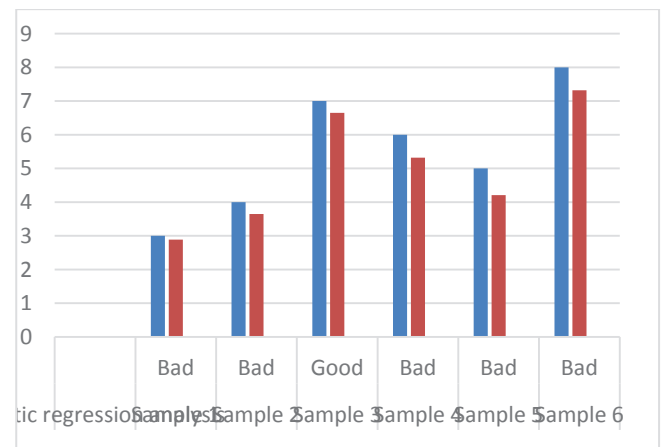


Fig. 6. Logistic Regression Prediction Analysis

The two models' logistic regression and SVM analysis when performed on training and testing dataset of Wine has got the values of standard regression coefficient, standard error and p value, when compared with both the values SVM holds the efficient results as compared to logistic regression illustrated in Figure 5 and Figure 6. From result analysis it's understood that values of unstable acidity and residue sugar does not affect quality of Wine, hence those can be eliminated from analysis. OQV represents the original quality value, QOR- Quality output for selected features for Wine, QOW- Quality output for selected features for Wine using SVM and logistic regression, as given in Table 7.

TABLE VII. SVM AND LOGISTIC REGRESSION PREDICTIONS' QUALITY

S A M P L E	OQV	SVM analysis		Logistic regression analysis	
		QOR	QOW	QOR	QOW
1	3.00	2.98	2.69	2.43	2.89
2	4.00	3.98	3.489	3.99	3.65
3	7.00	6.45	6.88	6.89	6.65
4	6.00	4.78	5.88	5.67	5.32
5	5.00	4.98	4.90	4.89	4.21
6	8.000	5.98	7.99	6.09	7.32

The figure 7 shows the quality versus residual sugar. The bar plot in figure 9 describes that the citric acid is directly proportional to the quality of Wine. As the quality of Wine rises the amount of citrus acid also increases finally this confirms that citrus acid is basic feature upon that quality of Wine is dependent as shown in figure 9. Free Sulphur dioxide is significantly contributing to the quality of Wine, the bar plot in figure 8 gives comparison of the free Sulphur dioxide values versus quality comparison.

The bar plot in the figure 5,6 illustrates the percentage of good quality Wine out of collected samples. Plot results convey that 80% of the data is classified as bad quality Wine and remaining 20% of Wine is good quality.

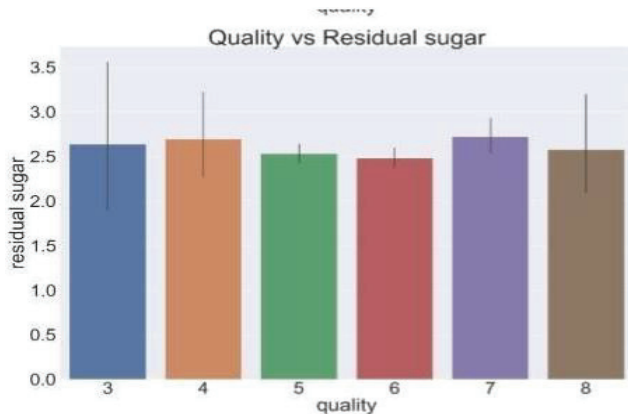


Fig. 7. Quality Vs Residual Sugar

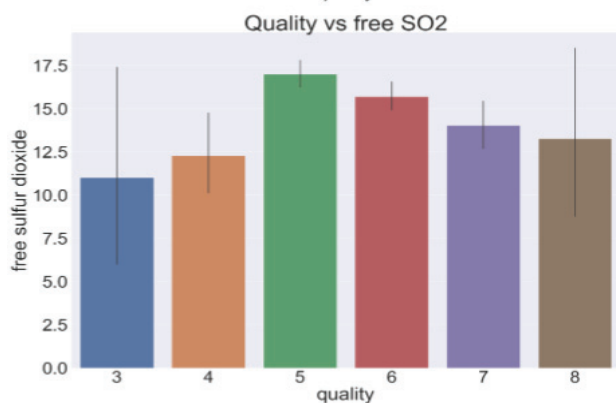


Fig. 8. Quality Vs Free SO₂

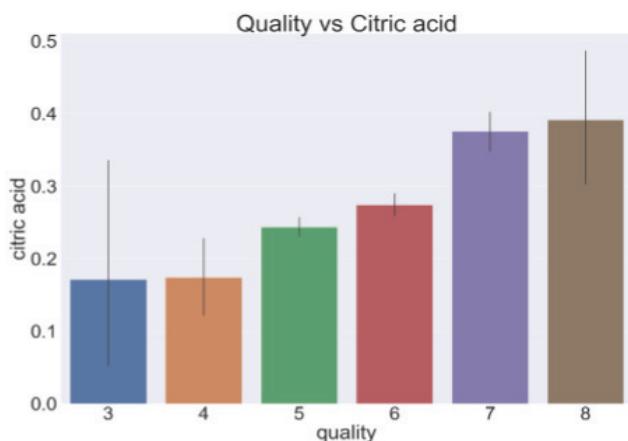


Fig. 9. Quality Vs Citric Acid

V. CONCLUSION

The work carried out involves deployment of three Machine Learning Algorithms, namely, Neural Network, SVM and Logistic Regression for prediction of Wine quality. The standard error SVM performed better than other methods with value of 0.00334. The P value for SVM is 0.0607 which is better as compared to logistic regression method. The experiment illustrates that the values of dependent variable such as volatile acidity, residual sugar, SO₂ and citric acid can be predicted more accurately when only significant factors are measured in prediction rather than measuring all factors.

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